

Global case study of digital marketing on social media by a top soda brand

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Summary

The Big Soda segment is central in modern food systems, and they invest heavily in digital marketing, but little is known about it. We aimed to analyze the digital marketing on Facebook of the soda brand with a major worldwide reach. Countries were described in terms of soda brand presence and popularity on Facebook according to countries' socio-demographic index (SDI) and the market share of soda brand (%MS). From 149 countries, 57.0% had soda brand's Facebook page among the top five in the number of followers within the beverage segment. Among them, digital marketing was described by the number of the page followers, the number of posts, and the number of interactions (likes, comments and shares) that each post received by Facebook users. Also, we analyzed the characteristics, and the use of marketing strategies on posts in a random sample of 10% ($n = 1217$) of all posts. We found that soda brand's popularity on Facebook was higher among countries with higher SDI and higher %MS. Also, the number of users that engaged with soda brand's posts was higher among countries with lower SDI and higher brand's %MS. The brand's posts focus on young adults, display of brand's products and glorification of its attributes. Other identified features include sport-related thematic and appeals to socialization and healthy eating. In addition, posts' characteristics differed with countries' characteristics. Soda brand digital marketing varies according to countries' characterization and maintains the brand position on the global marketing of beverages.

Lay summary

The concentration of power of transnationals food and beverage corporations across the entire food system chain has contributed to unfavorable outcomes in the health and diets of the population. The reasons may be due to the low-nutritional value of the foods and beverages produced by these transnationals and the wide promotion of these products through marketing strategies. These two points are important stages of modern food systems. The marketing practices of these companies influence the food choices of individuals. They are attracted by the high level of persuasion employed through different strategies, especially in the digital environment. Strategies in the digital environment are widely available and accessed by individuals in every country in the world. For this reason, we focus on understanding the digital marketing of an important soda brand with a major worldwide reach to understand its particulars. We show that this top big soda brand uses digital marketing to promote its products globally and presenting particularities aligned with the socio-demographic conditions of the countries and regarding its market share. These results show that monitoring the behavior, especially on digital media, of large food and beverage companies is essential to advance policies that regulate market practices to protect individuals from their externalities in the field of global public health.

Keywords: carbonated beverages, social media, marketing, commerce

INTRODUCTION

The modern food systems, dominated by transnational food and non-alcoholic beverage industries (such as soft drink companies known as Big Soda),

is a central determinant of the global syndemic of obesity, undernutrition and climate change (Popkin and Reardon, 2018; Swinburn *et al.*, 2019). Soft drinks are a meaningful piece of this process, as their

production and consumption are spread worldwide. They are rich in added sugars while containing low levels of vitamins and minerals (Vartanian *et al.*, 2007). The excessive consumption of these beverages has been associated with overweight (Scharf and DeBoer, 2016; González-Morales *et al.*, 2020), metabolic syndrome (Narain *et al.*, 2017) and cardiovascular diseases (Narain *et al.*, 2016). In addition, they are also responsible for high levels of greenhouse gas emissions (Kamp *et al.*, 2018).

Food marketing is an important determinant of food consumption (Vukmirovic, 2015; Boyland *et al.*, 2016; Russell *et al.*, 2019). In order to increase food and beverage sales all over the world, transnational food and non-alcoholic beverage industries can adopt different formats of food marketing according to the country's food marketing regulatory code and food culture. These formats can also be tailored to different socioeconomic population groups (Clapp and Scrinis, 2017; Baker *et al.*, 2020). Although food marketing is widespread in all mass media, the digital environment is the current big thing. Given the high levels of engagement of individuals with the digital food environment and the many possibilities of immersive and interactive strategies on these media, the ultra-processed food industries have considerably expanded their food promotion activities on social media platforms (UNSCN, 2020).

Some studies have already monitored food marketing in the digital food environment (Freeman *et al.*, 2014; Boelsen-Robinson *et al.*, 2016; Brownbill *et al.*, 2018; Buchanan *et al.*, 2018; Horta *et al.*, 2018; Vandevijvere *et al.*, 2018; Vassallo *et al.*, 2018; Jaichuen *et al.*, 2019; Bragg *et al.*, 2020). However, until now, only one study has compared the use of marketing strategies in this media (specifically on websites) by ultra-processed food industries in six countries according to their economic development (Bragg *et al.*, 2017). The understanding of the digital marketing of ultra-processed food over the world provides evidence about the harmful activities of these industries in different countries which can guide public policies formulation aiming to protect people from unhealthy food marketing.

To provide such evidence, food marketing monitoring can benefit from choosing a brand with worldwide reach and a digital platform highly accessed by individuals. In the segment of Big Soda, the brand with a major worldwide reach is present in >200 countries and territories (Coca-Cola, 2018) and has high investments in marketing that summed US\$ 4 billion per year in the last 6 years (Statista, 2020a). Also, currently, Facebook is the biggest digital platform worldwide with 2.7 billion users (about 36.0% of the world population) making it the most relevant scenario for the case study (Statista, 2020b).

The previously mentioned top soda brand has pages on Facebook in different countries that reach together >107 million followers.

Then, this study aims to analyze the digital marketing of a top big soda brand (from now on referred as 'brand') on Facebook over the world according to countries socio-demographic development and the brand market share.

METHODS

Sample definition

The Socialbakers platform (www.socialbakers.com) was consulted to construct the study sample. The filters of search 'brands in the beverage sector' were used (which includes coffee, soft drinks, water, teas and energy drinks). Between 1 July and 10 July 2019, 149 countries (77.2% of all countries in the world) were listed on the platform and only 30 (20.2%) had no brand page on Facebook. Among the others ($n = 119$), for 85 (57%) countries, the brand was popular on Facebook as these pages were among the top five beverages brand according to the number of Facebook followers. The other 34 (22.8%) countries had brand pages on Facebook, but these pages were not among the top five pages for beverages brands, and these countries were considered brand less-popular on Facebook (Figure 1).

Countries characterization

The 2017 Socio-demographic Index (SDI), developed by the Institute for Health Metrics and Evaluation, was used to characterize the studied countries in terms of socio-demographic development. It is a composite index that includes per capita income, average schooling and the total fertility rate. The index expresses a value between 0 and 1; values near 0 indicating a lower level of development based on the measured parameters (IHME, 2020). The SDI information was available for 137 countries and, among them, 34.3% has low or low-middle socio-demographic development (SDI = 0.00–0.63); 24.8% middle development (SDI = 0.64–0.74) and 40.9% high-middle or high development (SDI = 0.75–1.00) (GBD, 2018). The distribution of the SDI by country can be found in the Supplemental material.

The brand market share (the percentage of off-trade volume sales of the studied brand within the soft drinks segment—%MS) in 2018, derived from the Euromonitor International Passport Global Market Information Database, was used to describe the brand presence in the market of the countries included in the study (<https://www.euromonitor.com/>). This information (%MS) was available for 95 countries. For almost half of them (48.4%), the brand has >20% of %MS and among 18.9% of the countries, the brand has



Fig. 1: Distribution of the countries according to the brand presence and popularity on Facebook. 2019. Note: Countries classified as 'absent' were those that did not had brand page on Facebook between 1 July and 10 July 2019. The category 'data not available' refers to countries not listed on the Socialbakers platform.

<10.0% of %MS. The distribution of the brand %MS by country can be found in the [Supplemental material](#).

Brand digital marketing on Facebook pages among popular countries

Among brand popular countries on Facebook ($n = 85$), digital marketing was described in terms of the number of the page followers, the number of posts and the number of interactions (likes, comments and shares) that each post received by Facebook users. This information referred to 2018 and, except for the number of posts, all the other variables were standardized by the number of Facebook users in each country to correct this information for the variability on the number of population and the number of Facebook users per country.

In addition, we analyzed the characteristics, and the use of marketing strategies on posts in a random sample of 10% ($n = 1217$) of all posts published by the brand in its official pages on Facebook in 2018 in each country. Such posts have the potential to reach both followers and non-followers of the brand through messages posted by the brand itself or by other Facebook users. The data collection protocol was developed specifically

for the study and included variables related to the posts' characteristics (type of post, display of brand products, presence of individuals, age of individuals and brand logo) and marketing strategies use (promotional characters, premium offers and claims) ([Box 1](#)). The data collection tool was based on the International Network for Food and Obesity/Non-communicable Diseases Research, Monitoring and Action Support food television monitoring protocol (<https://www.informas.org/resources/>) and on previous studies that have monitored food marketing on social media ([Freeman et al., 2014](#); [Brownbill et al., 2018](#); [Bragg et al., 2020](#)). Double data entry was conducted by trained researchers, and all divergences were discussed and solved by them.

Statistical analysis

First, we tested the correlation between SDI and %MS among the countries that had both information available ($n = 95$), by applying Spearman's rank-order correlation. Second, SDI and %MS were described in median and percentiles 25 and 75 in the whole sample and accordance with the brand presence and popularity on Facebook. Third, brand

Box 1: Description of digital marketing variables identified in the posts

Variables	Description	Example
Characteristics of posts		
Post type	Nature of post: photo or video/gif	—
Display of brand products	Display of products of the brand in posts	 The image shows a branded product in evidence
Presence of individuals	Presence of people in posts	 The image shows people holding a branded product
Age of individuals	The age range of individuals in posts (children or adolescents, young adults and adults), when applicable	 The image shows young people drinking a branded product
Brand logo	Exhibition of the brand logo	 The image shows the company's logo

Variables	Description	Example
Marketing strategies—promotional characters		
Mention commemorative dates	Mention of commemorative dates such as Christmas, New Years' Eve, Easter, etc	 The image shows the association of a branded product with a Christmas character
Child directed elements (including child presence)	Presence of children or childish elements such as dolls or toys	 The image shows the presence of adults with a child (i.e. family) indicating an indirect consumption of a branded product by the child
Sports celebrities	Representation of sports celebrities internationally known	 The image shows an internationally known former professional footballer (David Trezeguet) consuming a branded product
Sports attributes	Representation of sports elements such as ball, team leaders or sports uniforms	 The image shows the presence of sporting characters, such as a football, one player on the football field and a sentence that suggests an appeal for a sporting event organized by the brand

Variables	Description	Example
Sports events	Representation of sports events internationally known such as the Olympics Games and the FIFA World Cup	 The image shows Zabivaka, the official mascot of the 2018 FIFA World Cup, dressed in the official shirt of the event with the company's logo
Non-sports celebrities	Representation of celebrities internationally known	 The image shows Maluma, an internationally known Colombian singer, songwriter and actor
Marketing strategies—premium offers		
Gifts and collectibles	Offer promotion of gifts or collectibles	 The image shows the association between purchasing a branded product and the chance of winning a trip to Russia and watching the FIFA WorldCup
New product launch	New product launch of brand such one new flavor	 The image shows the launch of a new flavor of the brand's product

Variables	Description	Example
Marketing strategies—claims		
Healthy eating	Association of a product with healthy foods or meals	 The image shows an association of the brand's product with a predominantly vegetable meal
Reference to socialization	Association of a product with people gathered, or friendship or family	 The image shows an association of the brand's product with a moment of socialization and joy between two during a meal
Reference to sustainability	Association of a product with elements of the environment such as water, air or soil	 The image shows an association of the brand's product with a sustainability event (GreenBag Fest). The post uses the appeal that the act of recycling is cool

popular countries on Facebook were classified as higher SDI and lower SDI when they had values of SDI above and below this cut-off point, respectively. The same was applied in the classification of countries as higher %MS and lower %MS. Fourth, data related to the number of followers, posts and likes, comments and shares were described among the 85 popular countries and stratified by SDI, and %MS categories. Finally, Chi-square test was applied to test differences in the characteristics, and the use of marketing strategies on posts published by the brand on Facebook in popular countries ($n = 85$) according to SDI and %MS categories. Differences at the 5% level were considered significant. The statistical analyzes

have been conducted using statistical software Stata 12.0.

RESULTS

Among the 95 countries that had both SDI and %MS information available, these variables were not correlated ($r = -0.0379$; $p = 0.7153$) (data not shown).

Countries that had brand pages on Facebook had higher SDI than countries without brand pages on Facebook (0.71 vs 0.64, respectively). %MS is similar between countries regardless the presence of the brand pages on Facebook (present: 19.55% vs absent: 22.50%). Brand popular countries on Facebook had

higher SDI (0.75) and higher %MS (21.65%) in comparison to the less-popular countries (0.65; 10.75%, respectively) (Table 1).

Among brand popular countries on Facebook, the median number of followers was higher among countries with higher SDI. However, countries with lower SDI had more likes, comments and shares per 1000 Facebook subscribers in comparison to countries with higher SDI (Figure 2a–e).

All brand popular countries on Facebook ($n = 85$) have SDI available data: lower SDI ($n = 43$) and higher SDI ($n = 42$). In the lowest category the values ranged between 0.33 and 0.74 and in the highest category between 0.75 and 0.91.

As for %MS, brand popular countries on Facebook with higher %MS had greater median values of followers, posts and likes, comments and shares per 1000 Facebook subscribers compared to the countries with lower %MS (Figure 3a–e).

Missing %MS data for nine brand popular countries on Facebook: lower %MS ($n = 38$) and higher %MS ($n = 38$). In the lowest category the values ranged between 3.4 and 21.6% and in the highest category between 21.7 and 64.3%.

Regarding posts characterization, posts from brand popular countries on Facebook ($n = 85$) used photos (59.5%) more frequently than videos/gif (40.5%) and young adults were the most common group shown (67.9%). Brand products were displayed on 67.9% of the posts and brand logo on 74.9%. Among the marketing strategies categories, messages of socialization (38.5%), sports attributes (21.0%) and sports events (16.6%) were the most frequent. Messages of healthy eating were present in 10.0% of the posts (Table 2).

In addition, posts from countries with lower SDI showed more frequently the brand logo ($p < 0.001$), sports attributes ($p < 0.001$), sports events ($p < 0.001$), gifts and collectibles ($p = 0.020$) and messages of

healthy eating ($p = 0.014$) than posts from countries with higher SDI. In contrast, countries with higher SDI published more posts showing young adults ($p = 0.005$) and using socialization messages ($p = 0.006$). Among countries with higher %MS, display of brand product ($p = 0.023$) and individuals ($p = 0.012$) were more frequent in the posts in addition to the mention of commemorative dates ($p = 0.026$) (Table 2).

DISCUSSION

This study has shown, for the first time, characteristics of worldwide Big Soda's digital marketing. We found that countries with higher socio-demographic development and a higher percentage of the brand participation in the market often have more popular pages on Facebook than countries without these characteristics. Besides that, the marketing content of posts made by the company is different according to country characteristics. In general, the brand focuses on young adults, on the display of brand products, and on the exaltation of its attributes. Other features of the posts are thematic related to sport, appeals to socialization and healthy eating. By these actions, the brand maintains its position on the global marketing of beverages and its power within the modern food systems.

The focus on young adults is strategic because they are the largest group of consumers of soft drinks. An analysis of the global consumption of sugar-sweetened beverages in 2010, in 187 countries, showed higher average consumption of these products among men (1.0 servings/day) and women (0.9 servings/day) between 20 and 39 years, compared to individuals of both sexes between 40- and 59-year-old (0.5 servings/day) and 60-year-old or more (0.3 serving/day) (Singh et al., 2015). In addition, Facebook is mainly used by young adults that spend much of their time using this social media (Kidd et al., 2021). The focus on young adults is concerning due to the high prevalence of obesity in this group (Abarca-Gómez

Table 1: Socio-demographic Index (SDI) and brand market share (%MS) among countries in accordance with brand presence and popularity on Facebook, 2019

Data	N ^a	Brand presence on Facebook				Brand popularity on Facebook			
		Present		Absent		Popular		Less-popular	
		(n = 119)		(n = 30)		(n = 85)		(n = 34)	
		Median	P25; P75	Median	P25; P75	Median	P25; P75	Median	P25; P75
SDI	137	0.71	0.59; 0.84	0.64	0.53; 0.76	0.75	0.62; 0.85	0.65	0.48; 0.74
%MS	95	19.55	11.75; 27.10	22.50	9.30; 23.90	21.65	14.45; 28.65	10.75	6.05; 18.10

Note: P, percentile.

^aFrom the 149 countries, 95 have available data for both metric; 42 countries only have available data for SDI; all countries that have %MS information also have SDI available data; 12 countries do not have available data for both metrics.

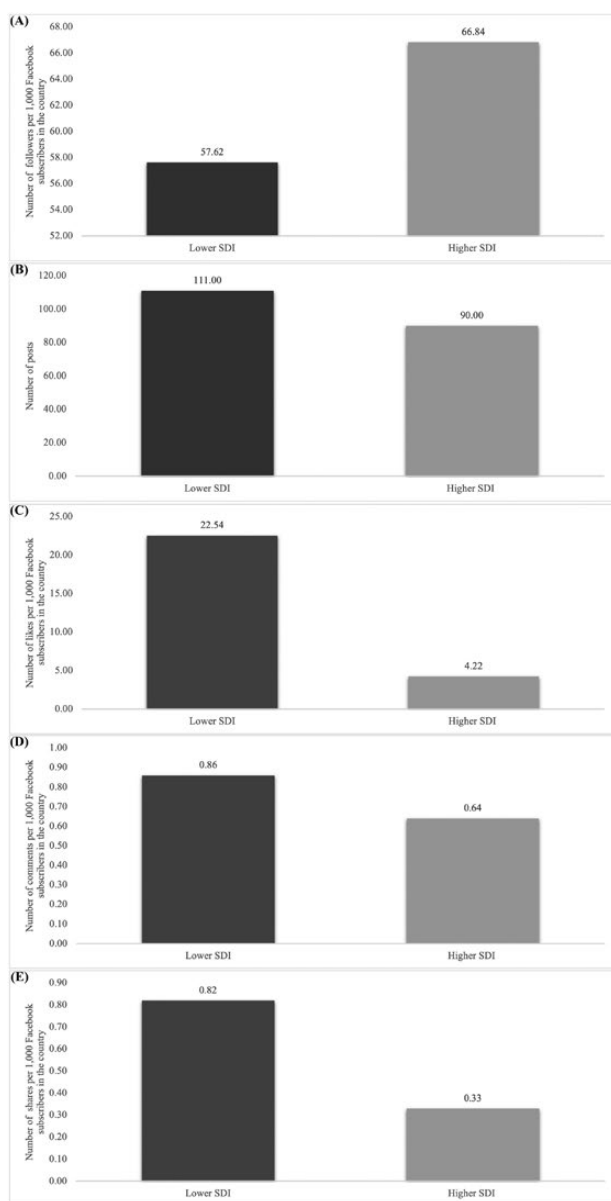


Fig. 2: Digital marketing (number of followers, posts and likes, shares and comments) among the brand popular countries on Facebook in accordance with the SDI. 2018.

et al., 2017) and their vulnerability to marketing from these companies. It may influence their feeling about the brand which might result in higher purchase intention and propensity to consume (Buchanan *et al.*, 2017).

As for the high presence of brand elements in advertisements, such as by displaying brand products, and the exaltation of its attribute, this is a pattern of ultra-processed advertising that has already been evidenced by other studies on social media (Freeman *et*

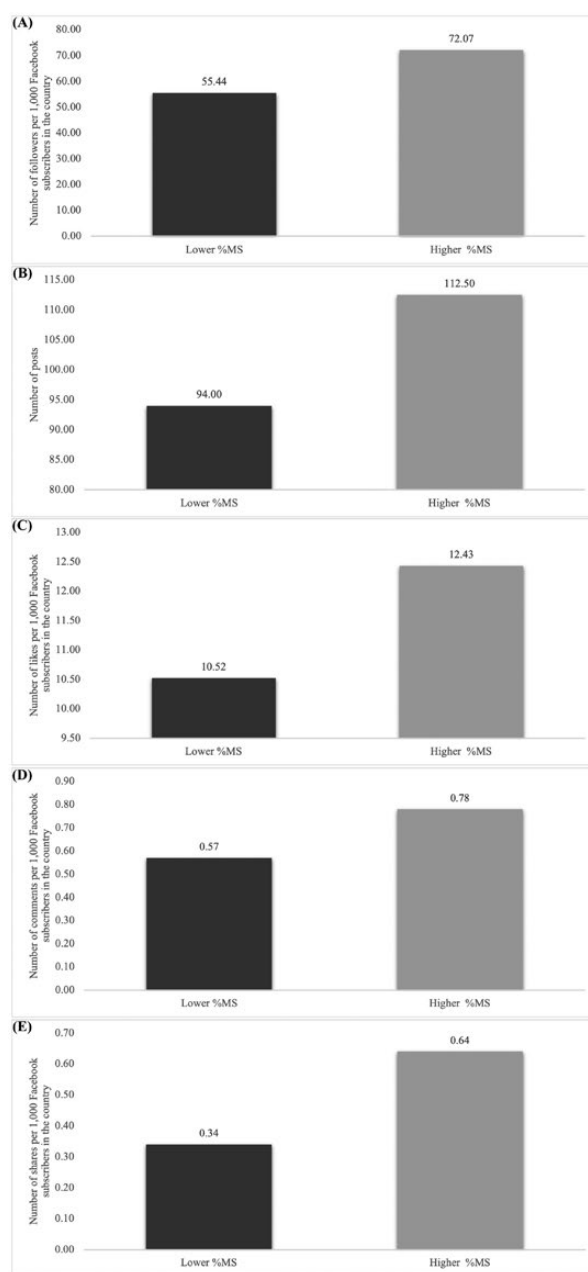


Fig. 3: Digital marketing (number of followers, posts and likes, shares and comments) among the brand popular countries on Facebook in accordance with the %MS. 2018.

al., 2014; Boelsen-Robinson *et al.*, 2016; Horta *et al.*, 2018; Vassallo *et al.*, 2018; Jaichuen *et al.*, 2019). In the studied case, these studies show that this technique is revealed in the advertisement through the colors adopted by the brand, the outline of the logo's writing and in its slogan (Boelsen-Robinson *et al.*, 2016; Vassallo *et al.*, 2018).

Table 2: Digital marketing (posts characterization and marketing strategy use) on Facebook pages among brand popular countries in accordance with the Socio-demographic Index (SDI) and brand market share (%MS) 2018

Variables	Total (%)	SDI (%)		<i>p</i> value	Brand %MS		<i>p</i> value
		Lower	Higher		Lower	Higher	
		(<i>n</i> = 43)	(<i>n</i> = 42)		(<i>n</i> = 38)	(<i>n</i> = 38)	
Characteristics of posts							
Post type							
Photo	59.5	59.1	60.1	0.720	57.6	54.5	0.316
Video/gif	40.5	40.9	39.9		42.4	45.5	
Presence of individuals							
No	37.9	37.9	37.8	0.956	42.1	34.5	0.012
Yes	62.1	62.1	62.2		57.9	65.5	
Age of individuals							
Only young adults	67.9	63.9	73.5	0.005	30.2	36.8	0.085
Other groups	32.1	36.1	26.5		69.8	63.2	
Display of brand products							
No	32.1	33.5	30.0	0.198	37.5	30.8	0.023
Yes	67.9	66.5	70.0		62.4	69.2	
Brand logo							
No	25.1	18.4	34.3	<0.001	25.8	26.7	0.728
Yes	74.9	81.6	65.7		74.2	73.3	
Marketing strategies—power of the advertising							
Mention of commemorative dates							
No	93.5	93.7	93.2	0.723	95.2	91.7	0.026
Yes	6.5	6.3	6.8		4.8	8.3	
Child directed elements (including child presence)							
No	95.4	95.2	95.7	0.629	96.5	94.0	0.063
Yes	4.6	4.8	4.3		3.5	6.0	
Sports celebrities							
No	99.8	99.7	100.0	0.225	99.7	99.8	0.882
Yes	0.2	0.3	0.0		0.2	0.1	
Sports attributes							
No	79.0	73.3	86.6	<0.001	77.7	79.1	0.592
Yes	21.0	26.7	13.4		22.3	20.9	
Sports events							
No	83.4	78.7	89.7	<0.001	81.1	85.1	0.078
Yes	16.6	21.3	10.3		19.0	14.9	
Non-sports celebrities							
No	98.3	97.7	99.0	0.082	98.9	97.3	0.073
Yes	1.7	2.3	0.1		1.1	2.6	
Marketing strategies—premium							
Gifts and collectibles							
No	94.3	93.0	96.1	0.020	93.7	95.0	0.340
Yes	5.7	7.0	3.9		6.3	5.0	
New product launch							
No	97.8	97.3	98.5	0.175	98.2	97.7	0.534
Yes	2.2	2.7	1.5		1.7	2.3	

Table 2. Continued

Variables	Total (%)	SDI (%)		<i>p</i> value	Brand %MS		<i>p</i> value
		Lower	Higher		Lower	Higher	
		(<i>n</i> = 43)	(<i>n</i> = 42)		(<i>n</i> = 38)	(<i>n</i> = 38)	
Marketing strategies—claims							
Healthy eating							
No	90.0	88.2	92.4	0.014	91.9	88.3	0.057
Yes	10.0	11.8	7.6		8.1	11.7	
Socialization							
No	61.5	64.8	57.0	0.006	61.6	61.1	0.868
Yes	38.5	35.2	43.0		38.4	38.9	
Sustainability							
No	99.4	99.4	99.4	0.980	99.6	99.3	0.572
Yes	0.6	0.6	0.6		0.4	0.7	

A total of 1217 posts was analyzed. All brand popular countries on Facebook ($n = 85$) have SDI available data and nine countries have missing data for %MS.

The high presence of sports content in posts can be explained by the fact that data collection was carried out during 2018 when the 21st edition of the FIFA Football World Cup in Russia was held. About 202 of the posts referred to sports events and 188 were about the World Cup. The top big soda brand is a brand that traditionally participates in the sponsorship of internationally recognized sporting events. The brand invests in advertising content with themes aligned to the competitions broadcasted on both digital and traditional media. Creation of exclusive and collectibles products and the organization of artistic events in the countries hosting the tournaments are also part of the strategy (Coca-Cola, 2020). These practices allow sponsoring companies to take advantage of the favoritism of certain teams in different sports to reach fans and influence the preference and purchase intentions of these consumers (Dixon *et al.*, 2019). The sponsorship of sporting events by the ultra-processed food industry is strongly criticized due to its potential to lead consumers to misrecognize the low healthiness of these products since the sports scenario is commonly associated with health promotion (Bragg *et al.*, 2018).

Socializing was also a frequent content in brand Facebook posts, especially among those involving young adults. Of the 469 posts with this theme, almost 60.0% featured this group. In Australia, a study showed exhaustive use of images of young adults in the brand Facebook posts, performing some leisure activity or with an appeal to friendship, fun and happiness (Brownbill *et al.*, 2018).

The theme healthy eating was also identified in a study carried out with 20 ultra-processed brands on the Instagram, Facebook, Twitter, Tumblr and Vine platforms (Bragg *et al.*, 2020). Two points must be highlighted in this context, the first concerns the brand's action to divert its role in current public health problems including obesity. The second point is the role that this practice plays in inducing the individual to associate unhealthy foods with healthy lifestyles.

Also, our results exposed meaningful particularities of the brand's digital marketing on Facebook. We found that countries with higher socio-demographic development and higher percentage of brand participation in the market often have more popular brand pages on Facebook than countries without these characteristics. Among popular countries, we observed a higher number of followers of the brand pages in (i) more developed countries, and (ii) with higher participation of the brand in the market. The number of posts and the volume of user interactions with the posts were higher in the least developed countries and in those countries where the brand has a higher market presence.

This scenario reflects the strategic investment in marketing made by the top big soda brand, which identifies the potential of its operations in different regions, with a focus on business consolidation and/or expansion (Clapp and Scrinis, 2017; Coca-Cola, 2018). In developing countries with intense urbanization processes encouraging the entry of transnational companies, ultra-processed companies (such as the studied brand) tend to invest in actions aiming to consolidate their brand (Popkin and Reardon, 2018). These actions

include: (i) a significant financial investment in product advertising, facilitated by the globalization of marketing agencies and social media technologies (Baker *et al.*, 2020); (ii) expansion of agricultural productivity of commodities with reduced input costs which favors the offer of products at lower prices; and (iii) innovation in food processing which increases its palatability and convenience (Popkin and Reardon, 2018).

Among the most developed countries, where the ultra-processed market is already consolidated, the companies focus on maintaining their position in the sales ranking and compete for participation in trade looking for opportunities to sell reformulated versions of their products with higher prices and higher aggregate value. For this, they invest in innovation, distribution and marketing strategies (Clapp and Scrinis, 2017).

POLICY IMPLICATIONS

We noticed that the content of the brand pages promotes unhealthy eating practices, which may be related to the inefficiency of the regulatory codes for the marketing of unhealthy foods adopted by most countries. Examples of such inefficiency include (i) centrality in self-regulatory policies in a model led by the food industry, which has proven to be very limited; (ii) the lack of a cross-border policy that minimizes differences in the application of guidelines, especially in neighboring countries with the same language or cultures; (iii) the centrality of regulatory policies aimed only at children. In the studied case, for example, the company has a responsible marketing global commitment that states no marketing is directed at children under 12 on all media. In our study, we found that this commitment holds, however, it does not consider the negative influence of the unhealthy food marketing exposure on other age groups. Consequently, the brand has been focusing its actions on the young adults' group and as discussed above, they are also at higher risk of consuming such beverages. In addition, the inefficiency of the regulatory codes for the marketing of unhealthy foods includes (iv) outdated guidelines that do not consider the particularities of digital media (Sacks and Looi, 2020; WCRFI, 2020; Coca-Cola, 2021).

Specifically, in digital media, researchers and policy makers face more pronounced barriers to the success of regulations than on the traditional media. To move forward on this agenda, the State must recognize its protective role; prioritize comprehensive national policies that protect the privacy of individuals' digital data; adapt and extend legislation, when existing, to the digital landscape; develop and collaborate with cross-border policies that consider the lack of barriers in the digital scenario; develop high-tech monitoring tools, as

well as appropriate sanctions and penalty mechanisms for both creators of marketing content and dissemination platforms. The food and beverages transnationals and the digital platforms companies must adapt their marketing practices to the regulations' norms; ensure transparency in the development of content and data use of digital users; disclose investment data in digital marketing as well as its expenses with content dissemination platforms, in addition to performance data and oversee the distribution of content prohibited by specific legislation (WHO EUROPE, 2016; UNSCN, 2020).

In addition, some actions with the focus on breaking the current political inertia are strongly recommended. Among the possible pathways, the following stand out: the decrease in the influence of major commercial interests in the process of developing public health protection policies and in the development of research focused on public health (Swinburn *et al.*, 2019; Vandevijvere and Kraak, 2019), which is a typical practice of the studied brand and other ultra-processed brands in different countries of the world (Serôdio *et al.*, 2018; Rey-López and Gonzalez, 2019; Mialon and Gomes, 2019; Serôdio *et al.*, 2020); the reorientation of the food industry in terms of its policies and civil liability; and the reinforcement of the systematic monitoring of marketing practices, as carried out by the present study, considering the local particularities, both related to development and market share of the industries (Swinburn *et al.*, 2019; Vandevijvere and Kraak, 2019).

LIMITATIONS

Despite the relevance and novelty of this study in identifying a top soda brand digital marketing across different countries on Facebook, some limitations need to be highlighted. First, the focus is on a single social media despite it being the most used worldwide (Statista, 2020b) and only one brand, although it is the main representative of the Big Soda segment worldwide (Coca-Cola, 2018). Second, it is also worth mentioning that some content may not have been identified due to the barriers to the interpretation of different languages. Finally, still related to the language barrier and added to the cultural, social and political differences of several countries analyzed, we were not able to identify the nature of the interactions, or the content generated by users about the evaluated brand.

CONCLUSION

The top big soda brand uses Facebook to promote its products globally. However, some differences in the digital marketing of the brand across countries were

noted according to the socio-demographic development and the brand market share.

Countries with higher SDI and higher %MS had Facebook pages of the brand more popular than countries with lower SDI and %MS. In addition, countries with higher SDI had more frequently Facebook pages of the brand than did the countries with lower SDI.

The number of posts and likes, comments and shares were also higher among brand popular countries on Facebook with lower SDI. In addition, brand Facebook pages from countries with higher participation of the brand on the market had more posts, followers and interactions than did the Facebook pages of the countries with lower %MS.

In general, brand posts focus on young adults, on display of brand products and on the exaltation of its attribute. Sports related thematic, socialization appeals and healthy lifestyle were also marketing content common on the posts. Besides that, characteristics of the posts and the use of marketing strategies differed in accordance with countries' characteristics.

Supplementary Material

Supplementary material is available at *Health Promotion International* online.

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Conflict of interest

The authors declare no conflict of interest.

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